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# A Novel Model Based on a Nonlinear Fractional System for Image Restoration

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**Abstract.** In this paper, we develop a nonlinear system involving the fractional Laplacian operator, based on the  $H^{-s}$ -norm and a decomposition approach. The proposed method integrates the advantages of the dynamical threshold-based Perona–Malik (DTPM) model in flat regions with the benefits of incorporating a weak norm in the fidelity term. This weak norm is particularly suited for capturing highly oscillatory features typically interpreted as texture. Using Schauder's fixed point theorem, we establish the existence and uniqueness of a weak solution to the system. To compute the solution numerically, we use a finite difference scheme to approximate the fractional Laplacian. Finally, we present comparative numerical experiments to evaluate the efficiency and effectiveness of the proposed model.

**Key words:** Image denoising, Nonlinear diffusion, Fractional-order PDE, Perona–Malik (PM) equation, Dynamical threshold.

**AMS subject classification:** .

## 1. Introduction

In this paper, we discuss the following reaction–diffusion system

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$$\left\{ \begin{array}{ll} \frac{\partial u}{\partial t} - \operatorname{div} \left( g(|\nabla u_{\sigma_1}|, |\nabla u_{\sigma_2}|) \nabla u \right) + 2\lambda w = 0 & \text{in } \Omega_T := \Omega \times (0, T), \\ \frac{\partial w}{\partial t} + (-\Delta)^s w + (f - u) = 0 & \text{in } \Omega_T := \Omega \times (0, T), \\ \langle \nabla u, \vec{n} \rangle = 0 & \text{on } \Sigma_T := \partial\Omega \times (0, T), \\ w = 0 & \text{in } \Omega_T^c := \mathbb{R}^N \setminus \overline{\Omega} \times (0, T), \\ u(0) = f, \quad w(0) = 0 & \text{in } \Omega, \end{array} \right. \quad (1.1)$$

where  $\Omega$  is a bounded domain of  $\mathbb{R}^N$  with an appropriately smooth boundary,  $n$  denotes the unit outer normal to  $\Omega$ ,  $\beta > 0$  and  $T > 0$ . Herein, we consider the fractional Laplacian on a bounded domain  $\Omega_T^c$  with extended homogeneous Dirichlet boundary conditions defined on  $\Omega_T^c$ . The diffusion function  $g(s)$  is a decreasing smooth function satisfying  $g(0) = 1$ ,  $g(s) > 0$ , and  $g(s) \rightarrow 0$  for  $s \rightarrow \infty$ ,

$$G_\sigma \in C^\infty(\mathbb{R}^N) \text{ is the smoothing kernel with } \int_{\mathbb{R}^N} G_\sigma(x) dx = 1.$$

Moreover  $\lim_{\sigma \rightarrow 0} G_\sigma = \delta_x$ , where  $\delta_x$  is the Dirac measure at point  $x$ ,  $f \in L^2(\Omega)$ . Also

$$u_\sigma := (G_\sigma * u)(x) = \int_{\mathbb{R}^N} G_\sigma(x - \xi) \tilde{u}(\xi) d\xi,$$

where  $\tilde{u}$  is an extension of  $u$  to  $\mathbb{R}^N$  (see [1]) and  $G_\sigma(x)$  is the Gaussian kernel, given by

$$G_\sigma(x) = \frac{1}{(\sqrt{2\pi}\sigma)^N} \exp\left(-\frac{\|x\|^2}{\sigma^2}\right), \quad x \in \mathbb{R}^N.$$

Moreover, different diffusion coefficients have been proposed. In current work, we adopt a new function  $g()$ , inspired by the Perona–Malik model [4] and further developed by Li et al. [3], to enhance the preservation of fine image structures. The diffusion function is defined as follows

$$g(|\nabla u_{\sigma_1}|, |\nabla u_{\sigma_2}|) = \frac{1}{1 + \left( \frac{|\nabla u_{\sigma_1}|}{K(|\nabla u_{\sigma_2}|)} \right)^2},$$

where the dynamical threshold function  $K(|\nabla u_{\sigma_2}|)$  is given as follows

$$K(|\nabla u_{\sigma_2}|) = \alpha(|\nabla u_{\sigma_2}|)k_1 + (1 - \alpha(|\nabla u_{\sigma_2}|))k_2,$$

and the dynamical weight  $\alpha(|\nabla u_{\sigma_2}|)$  is given by

$$\alpha(|\nabla u_{\sigma_2}|) = 1 - \frac{1}{1 + |\nabla u_{\sigma_2}|^2},$$

with  $k_1 < k_2$  representing the lower and upper threshold bounds, respectively.

## 2. Numerical Results

To compute numerically problem, we discretize it by finite difference method. On the other hand, we approximate the fractional Laplacian by the finite difference methods (see [2]).



Figure 1: \*  
PSNR=28.362

Figure 2: \*  
PSNR=27.380

Figure 3: \*  
PSNR=27.885

Figure 4: \*  
PSNR=29.463

Figure 5: First row: Image clear. Second row: Images noisy  $\sigma_0^2 = 0.05$ . Third row: Restored images using  $H^{-0.5}$ -norm



### 3. Conclusion

In this work, we present a novel approach for image denoising by combining a dynamical threshold-based Perona–Malik (DTPM) model with a nonlinear system involving the fractional Laplacian. This hybrid model effectively preserves edges, suppresses the staircase effect, and prevents the introduction of artificial features. The fractional Laplacian is derived from a weak norm defined in the dual of a fractional Sobolev space, which is particularly well-suited for capturing patterns, oscillations, and textures. This dual space also provides a close approximation to the Meyer norm, enhancing the model’s ability to handle complex image structures. Finally, we present experimental results that demonstrate the superior performance of our approach compared to existing methods in the literature, confirming its effectiveness in denoising tasks.

### References

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